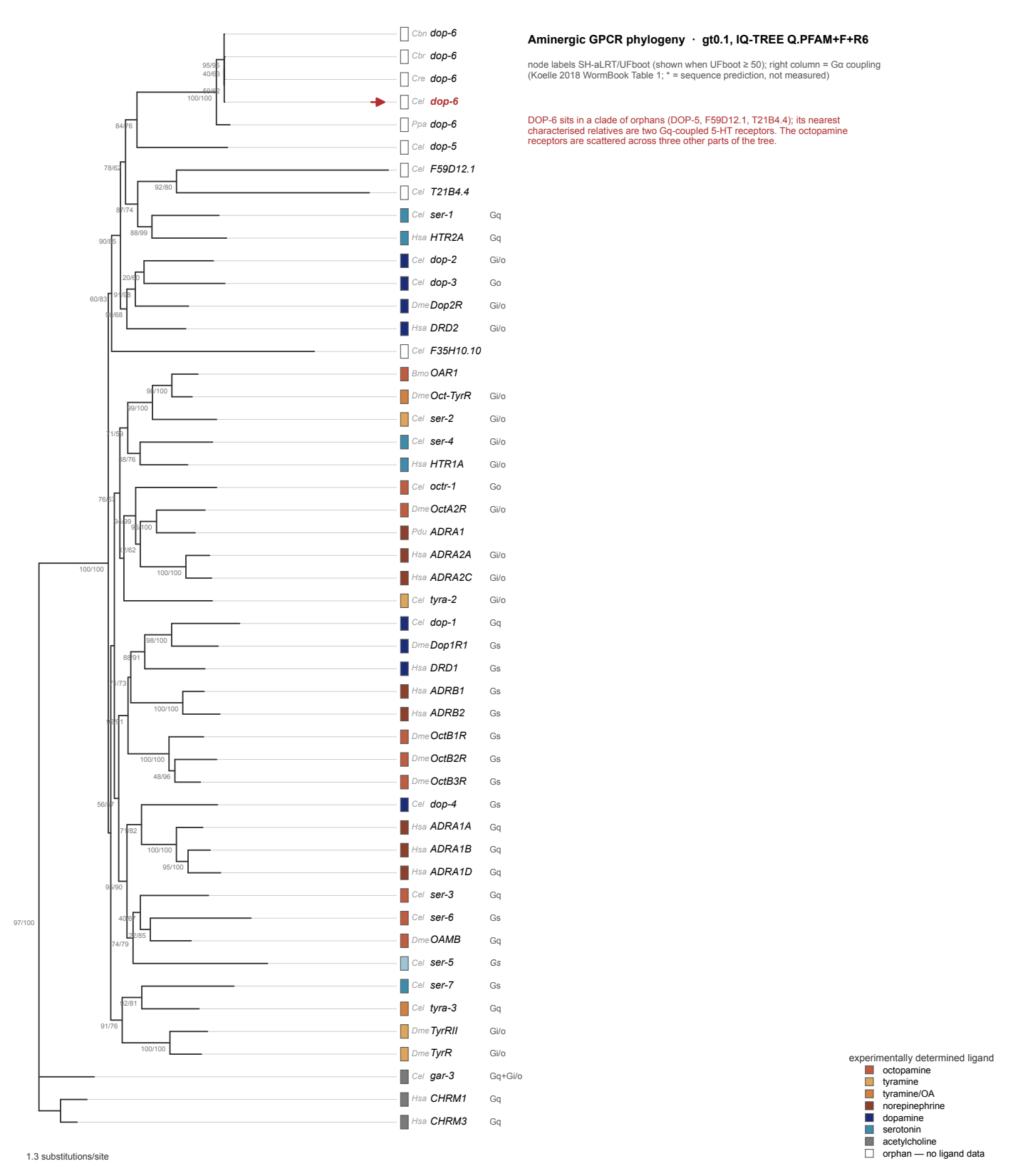

Aminergic GPCR phylogeny · gt0.1, IQ-TREE Q.PFAM+F+R6
node labels SH-aLRT/UFboot (shown when UFboot ≥ 50); right column = Gα coupling (Koelle 2018 WormBook Table 1; * = sequence prediction, not measured)
DOP-6 sits in a clade of orphans (DOP-5, F59D12.1, T21B4.4); its nearest characterised relatives are two Gq-coupled 5-HT receptors. The octopamine receptors are scattered across three other parts of the tree.
Cbn dop-6
Cbr dop-6
Cre dop-6
Cel dop-6
Ppa dop-6
Cel dop-5
Cel F59D12.1
Cel T21B4.4
Cel ser-1 Gq
Hsa HTR2A Gq
Cel dop-2 Gi/o
Cel dop-3 Go
Dme Dop2R Gi/o
Hsa DRD2 Gi/o
Cel F35H10.10
Bmo OAR1
Dme Oct-TyrR Gi/o
Cel ser-2 Gi/o
Cel ser-4 Gi/o
Hsa HTR1A Gi/o
Cel octr-1 Go
Dme OctA2R Gi/o
Pdu ADRA1
Hsa ADRA2A Gi/o
Hsa ADRA2C Gi/o
Cel tyra-2 Gi/o
Cel dop-1 Gq
Dme Dop1R1 Gs
Hsa DRD1 Gs
Hsa ADRB1 Gs
Hsa ADRB2 Gs
Dme OctB1R Gs
Dme OctB2R Gs
Dme OctB3R Gs
Cel dop-4 Gs
Hsa ADRA1A Gq
Hsa ADRA1B Gq
Hsa ADRA1D Gq
Cel ser-3 Gq
Cel ser-6 Gs
Dme OAMB Gq
Cel ser-5 Gs
Cel ser-7 Gs
Cel tyra-3 Gq
Dme TyrRII Gi/o
Dme TyrR Gi/o
Cel gar-3 Gq+Gi/o
Hsa CHRM1 Gq
Hsa CHRM3 Gq
1.3 substitutions/site
experimentally determined ligand
octopamine
tyramine
tyramine/OA
norepinephrine
dopamine
serotonin
acetylcholine
orphan — no ligand data